

Abderrahmane Benfanich

Ph.D. Candidate in Mathematics • University of Ottawa

Department of Mathematics and Statistics • Ottawa, ON, Canada

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RESEARCH INTERESTS

Numerical Analysis, Finite Element Methods (FEM), Nonlinear & Degenerate Partial Differential Equations, Porous Media Flow (Richards Equation), Structure-Preserving Discretizations, Discrete Maximum & Positivity Principles, Domain Decomposition Methods, Parallel Scientific Computing.

EDUCATION

University of Ottawa

09/2024 – Present

Ph.D. in Mathematics and Statistics

Ottawa, ON, Canada

- **Dissertation:** Numerical Analysis and Robust Finite Element Schemes for Nonlinear Porous Media Flow.
- **Focus:** Development of structure-preserving numerical schemes, auxiliary variable formulations, and domain decomposition solvers for infiltration dynamics.
- **Advisors:** Prof. Yves Bourgault and Prof. Abdelaziz Beljadid.

University of Ottawa

09/2023 – 09/2024

M.Sc. in Mathematics and Statistics

Ottawa, ON, Canada

- Accelerated and fast-tracked directly into the Ph.D. program following first-year research performance and academic excellence in graduate coursework.

Cadi Ayyad University

09/2020 – 06/2023

B.Sc. in Mathematics and Applications

Marrakech, Morocco

- Comprehensive curriculum covering pure analysis, linear algebra, numerical analysis, probability, and mathematical physics.

PUBLICATIONS & PREPRINTS

UNDER REVIEW Benfanich, A., Bourgault, Y., & Beljadid, A. (2026). *Discrete positivity and maximum principles for a finite element discretization of the Richards equation*. Under review in **Journal of Computation and Applied Mathematics**.

UNDER REVIEW Benfanich, A., Bourgault, Y., & Beljadid, A. (2026). *Mathematical analysis for a doubly degenerate parabolic equation: Application to the Richards equation*. arXiv:2602.19037. Under review in **IMA Journal of Numerical Analysis**.

ACCEPTED Benfanich, A., Bourgault, Y., & Beljadid, A. (2026). *A finite element method using a bounded auxiliary variable for solving the Richards equation*. Accepted, **Journal of Computational Physics**.

AWARDS, HONORS & FELLOWSHIPS

InterPore Journal Invited Student Paper Award Nomination

06/2026

Nominated by session chairs at InterPore 2026 for outstanding presentation and contributions to porous media modeling.

International Doctoral Scholarship

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University of Ottawa

09/2024 – Present

Awarded for doctoral academic standing and research merit.

Ph.D. Admission Scholarship

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University of Ottawa

09/2024 – Present

Supervisor's Research Scholarship

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University of Ottawa

09/2023 – 08/2024

CONFERENCE PRESENTATIONS & TALKS

WCCM-ECCOMAS 2026 — World Congress on Computational Mechanics

07/2026

Oral Presentation: "A Robust, Structure-Preserving Finite Element Solver for the Richards Equation using a Bounded Auxiliary Variable and Domain Decomposition." Munich, Germany.

InterPore 2026 — 18th International Conference on Porous Media

05/2026

Oral Presentation: "A Bounded Auxiliary Variable Method for Infiltration Modeling and Domain Decomposition." Nantes, France.

SIAM / CAIMS Annual Meeting (AN25) — Joint Annual Meeting

07/2025

Oral Presentation: "A Finite Element Method for Solving the Richards Equation Using a New Bounded Auxiliary Variable." Minisymposium MS138: *New Directions in Finite Element Methods Part III*. Montréal, QC, Canada.

TEACHING & ACADEMIC SUPPORT EXPERIENCE

Graduate Teaching Assistant & Guest Lecturer

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University of Ottawa

09/2023 – Present

- **Guest Lecturer — MAT 3720:** Delivered substitute university course lectures in French on ordinary differential equations and numerical methods for regular faculty (Prof. Arian Novruzli).
- **Discussion Group (DGD) Instructor (Taught in French; selected courses include):**
 - **MAT 1748:** *Mathématiques discrètes pour l'informatique* – Logic, set theory, induction, graph theory, and combinatorics.
 - **MAT 1739:** *Introduction au calcul et vecteurs* – Differential calculus techniques, coordinate systems, and vector analysis.
 - **MAT 1720 / MAT 1722:** *Calcul différentiel et intégral I & II* – Limits, continuity, integration techniques, and series convergence.
 - **MAT 1741:** *Introduction à l'algèbre linéaire* – Matrix operations, vector spaces, eigenvalues, and linear transformations.
- **Course Grader & Evaluator (Selected courses include):**
 - **MAT 3748:** *Algèbre appliquée / Applied Algebra* (3rd-year) – Graded group theory and finite field examinations.
 - **MAT 2777:** *Probabilités et statistique pour les sciences* (2nd-year) – Graded continuous distributions, hypothesis testing, and regression problem sets.
 - Comprehensive evaluation of assignments, midterm exams, and final exams across core undergraduate STEM courses.

Math Help Center Tutor

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University of Ottawa

09/2023 – Present

Provided individualized drop-in academic tutoring and proof-writing guidance for undergraduate students across foundational pure and applied mathematics courses in both English and French.

COMPUTATIONAL & TECHNICAL SKILLS

Scientific Computing:	Finite Element Methods (FEM), FreeFEM++, NumPy, SciPy, ParaView, High-Performance Computing (HPC)
Programming & Tools:	Python, C++, LaTeX, Git, Visual Studio Code, Linux/Bash, Xournal++
Languages:	English (Fluent), French (Fluent), Arabic (Native)

ACADEMIC REFERENCES

Prof. Yves Bourgault (Ph.D. Advisor)
Professor, Dept. of Mathematics & Statistics
University of Ottawa
Email: yvesbourgault@uottawa.ca

Prof. Abdelaziz Beljadid (Ph.D. Co-Advisor)
Associate Professor, Dept. of Mathematics & Statistics
University of Ottawa
Email: abeljadi@uottawa.ca